nleFlow plan

nleFlow: plan and recommendation

Planning stage for AH, 2026-10-16, from `handover/NLEFLOW_PLAN_PROMPT.md`. Branch `nleflow-plan`. Everything stated about the code was read from the files cited; everything stated about what runs was run on this cloud instance (Ubuntu 24.04, 4 cores, 15 GB, R 4.3.3) and is reported in §9.5 with what was not run.

0. Summary and recommendation

Question. Is it worth building a separate R package, `nleFlow`, that (1) learns a new neural likelihood flow by our recipe, (2) exports flows and LANs to EMC2 cards and to ONNX, and (3) converts ONNX LANs to EMC2 cards?

Recommendation: build a smaller thing, in two halves, and no R package called nleFlow.

1. **A Python package `nleflow`** for the parts that are Python by nature: a model specification, one training command, the checkpoint exporter with golden vectors and the float64 gate, the ONNX exporter (including the DDMnle graph that does not exist yet), and the ONNX-to-card converter (today an EMC2 `inst/scripts` file that needs Python anyway). This is where "learn a new flow" lives, and it is a generalisation of `handover/nle-methods/flow/`, not a rewrite.
2. **The R side stays in EMC2 `dev-nle`**, which already holds the card loader for four kinds of network, `register_nn_model()`, the validation kit (`nn_cell`, `nn_in_box`, `nn_total_mass`, `nn_score_bias`, `nn_posterior_shift`, `nn_sbc_cell`), the converter, a vignette and 500+ passing tests (§1.3). What EMC2 lacks is small and specific: a versioned, written card spec; the DDMnle (scaled) kind wired in R, not only in C++; a golden-vector check; a fix for R < 4.5 (§1.3). The nle repo's R reference evaluators (`port/R/flow_race.R`, `flow_ddm.R`, `ddm_scaled.R`) become the fixtures EMC2's tests pin against, as they already are for two of the three.

Why not an R package that trains. Training is JAX. Wrapping it in R through reticulate adds the heaviest dependency in the project (a pinned JAX environment, GPU optional) to an R package that could not go to CRAN and would duplicate EMC2's card loader, which is the one piece that must not drift. An R-only package that only exports, converts and validates would duplicate what `dev-nle` has already, with tests, and would re-create the "two copies of the same evaluator" trap that cost this project weeks (`handover/EMC2_IMPLEMENTATION_BRIEF.md:15-19`).

Why a Python package rather than the scripts as they are. Nothing in `flow/` is general: every context list, transform, box, zero-box constant and output path is a literal in a script (§4.2, §8). A user with a new model edits six files. The two worked examples, DDMnle and the RDM, were each built by adding a flag and a context list to those files; a third model would be a third flag. A model specification plus one command is the smallest change that makes the recipe usable by someone else, and it is also what makes the two examples reproducible as tests.

Audience, honestly. The users are us first (Julian, Malte, AH: the paper's reproducibility and the next models), EMC2 users with an intractable model second (a handful of labs), and HSSM users wanting EMC2 third (they need only the converter, which exists). That audience justifies a lean package with two worked examples and a verification command, not a product.

Smallest useful first milestone (about a week, §11): the card spec v1 written down and versioned; the DDMnle kind wired in R inside EMC2 (today reachable only through `port/R/ddm_scaled.R` plus a scaffold card that is not committed); `nn_golden_check()`; golden vectors regenerated for the committed DDMnle card (its golden file does not exist, §2.4); and one verification command that passes on a fresh machine. Nothing in that milestone depends on deciding the Python package.

Decisions AH needs to make are listed in §11.3.

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1. What exists today, verified here

1.1 EMC2 `dev-nle` (the tarball, 142bd497)

`handover/emc2/emc2_142bd497.tar.gz` is `git archive` of `dev-nle` at 142bd497, `Version: 3.4.1.9001` (`DESCRIPTION:3` of the extracted tree; `handover/emc2/README.md:4` says "3.4.1", which was the dev-nn era build). It is the final sampler (`dev-sampler` aa66e11d merged into `dev-nle`, `handover/MALTE_STAGES.md:1106-1108`).

It is less "not on GitHub" than the brief says. `ampl-psych/EMC2` has a `dev-nle` branch at 154f21ad (2026-10-02, the `qbinom` fix) on top of 14c12c79 (2026-09-28, natural-scale race flows, hybrid race, CRDM) and d49d190b (zero-box). So the whole NN layer (cards, `register_nn_model`, the kit, the converter, the tests) is public. What is not pushed is the merge with the final sampler (142bd497) and the scaled-DDM patch (`handover/emc2/emc2_dev-nle-scaled.patch`). Meanwhile `dev` on GitHub has the new sampler (8b9bdbf, 2026-10-06) and later commits, so `dev-nle` is now behind `dev` on the sampler and ahead of it on NN: a rebase is due before any of this work lands.

What the tree contains (file:line in the extracted EMC2):

piece	where	notes
card kinds	`R/nn_register.R:55` `nn_kinds <- c("flow_joint","flow_race","regression_joint","mlp_joint")`	inferred from the card (`:208-213`): `flow_mlp+classifier_mlp` → flow_joint; `mlp+spline` → flow_race; `mlp` with `kind == "mlp_joint"` → LAN; other `mlp` → regression
`register_nn_model(path, pars, transforms, kind, cdf, exceptions, pre, twin, p_types, hybrid)`	`R/nn_register.R:503-505`, roxygen `:240-502`	`.rds`, `.json` (needs jsonlite) or a shipped name; context order asserted against the card (`:521-536`); transforms must agree (`:537-559`); every twin parameter must be an input, used by `pre`, or fixed (`:638-641`)
transforms	`:57` `identity, log, probit, zerobox`; zerobox = asinh(x/c) with `zerobox_c` per input (`:66-93`)	
compiled evaluators	`src/model_NN.h` (dispatch, `NnSpec` `:39-60`), `src/flow_ddm.cpp`, `src/flow_race.cpp`, `src/mlp_lik.cpp`, `src/nle_flow.h`	refuses censoring and truncation (`model_NN.h:96-111`); joint kinds are two-response (`:165-169`)
validation kit	`R/nn_validate.R`: `nn_cell` `:94`, `nn_in_box` `:326`, `nn_total_mass` `:621`, `nn_score_bias` `:734`, `nn_posterior_shift` `:876`, `nn_sbc_cell` `:1047`, `nn_sbc_summary` `:1101`	`run_sbc` refuses a prior that leaves the box (`R/SBC.R:34`)
converters	`inst/scripts/onnx_to_card.py` (MLP LANs, tanh only, rank-1 input, one preset `ddm_uniform_st`), `inst/scripts/regress_to_card.R`	
gates and scripts	`inst/scripts/nle_acceptance.R`, `nle_validation_kit.R`	the acceptance gate compares shipped artefacts with the NLE R port to 1e-10
docs	`man/register_nn_model.Rd` (the only description of the card fields, roxygen `:263-379`), `vignettes/neural-likelihoods.Rmd`	no standalone card spec, no version field
shipped artefacts	`inst/extdata/flownn/`: `ddm_cap256w_c4.rds`, `ddm_st0zero.rds`, `rdm_small.rds`, `MANIFEST` (sha256)	
tests	`tests/testthat/test-nn-{port,register,mlp,zerobox,malte,native,validate,reg-`	

piece	where	notes
	s402}.R; golden files under tests/testthat/golden/ include ddm_uniform_st.onnx, lan_ddm_uniform_st_ort.csv, rdm_rdm_small_e195_golden.csv, ddm_cap256w_golden.csv	

The scaled patch (handover/emc2/emc2_dev-nle-scaled.patch, 177 lines, src/model_NN.h, src/flow_ddm.cpp, src/nle_native.h) adds NnSpec::scaled, the eight DDM columns, nn_trial_ll_scaled() and nle_ddm_native_full() (pdf and CDF). **It is C++ only**: nothing in R/ sets scaled or sc_pars. The R-side wiring is ddm_sc_native_reg() in handover/nle-methods/port/R/ddm_scaled.R:115-135, which copies a DDM registration built from a zero-box "scaffold" card (sbc/scaled_cell.R:27,38: ddm_zb_s903_e271_weights.json) and rewrites its native spec. That scaffold card is not in the repository (find returns only .out/.rds under sbc/zerobox_postshift_v1/). See §2.4.

1.2 The Python training and export code

handover/nle-methods/flow/ (JAX 0.5.3, flax 0.10.6, distrax 0.1.5, optax 0.2.8, orbax < 0.11.2, numpy 2.2.6, Python 3.11–3.12: pyproject.toml:10-28, uv.lock). onnx and onnxruntime are imported by scripts/export_onnx_ddm.py and scripts/onnx_check.py but are **not in the lock or pyproject**. The handover copy and /home/user/nle/flow have diverged in both directions: the handover copy has --zerobox, --scaled and the float64 gate; the root copy has --decision-time for the DDM, a README, requirements.txt, notebooks and tests (diff -rq; details in §8).

There is no generic entry point. The closest are scripts/train_race.py with its CONTEXT dict (:71-74: RDM v,B,t0,s,A; LNR m,s,t0) and scripts/export_flow.py with MODEL_REGISTRY (:74-94); the DDM has its own train_ddm.py, retrain_classifier.py, export_flow_ddm.py, each with a CONTEXT* literal per build (export_flow_ddm.py:45-61).

1.3 What I ran here (fresh Ubuntu 24.04 container, R 4.3.3)

step	result
R 4.3.3 from apt	45 s
16 EMC2 dependencies as r-cran-* from apt	2 min 16 s; WienR is not packaged, built from the GitHub CRAN mirror in 18 s (CRAN itself is blocked by this environment's network policy)
EMC2 dev-nle 142bd497 from the tarball, dedicated library	builds; packageVersion 3.4.1.9001; patched copy with --preclean -j4: 78 s
Python stack by uv (jax 0.5.3, flax 0.10.6, onnx 1.23.2, onnxruntime 1.31.0, numpy 2.2.6)	11 s
port/tests/test_ddm_scaled.R (DDMsc oracle: rescaling, t0 shift, st0 convolution vs dDDM)	**PASS**, 7 s (max
flow/scripts/onnx_check.py on ddm_cap256w_c4_shared.onnx and ddm_cap256w_c4.onnx against 400 R-port cases	max
inst/scripts/onnx_to_card.py on the LAN ddm_uniform_st.onnx from EMC2's test tree, --preset ddm_uniform_st --verify 2000	writes the card; card vs onnxruntime max
EMC2's own nn tests (test-nn-port/register/mlp/zerobox/malte/native)	**503 expectations pass, 10 errors, 3 skips**; every error is one trap: tools::sha256sum does not exist before R 4.5, nle_file_hash (R/nn_register.R:42-46) falls back to md5, and inst/extdata/flownn/MANIFEST holds sha256 only, so every *shipped* artefact fails "does not match the sha256 pinned". JSON cards register fine. On R ≥ 4.5 (the laptop runs 4.6.0) this never shows
the committed DDMnle card port/export/ddm_sc_s1106_e245_weights.json through ddm_sc_card_evaluator()	loads; 0.415 ms per 400-row evaluator call; total mass by quadrature up to log u = 2 is 0.9757 and the sum of the two defective CDFs at the spline's range_max is exactly Φ(2) = 0.97725 for every θ tried (§2.4)
the compiled DDMnle path: patched build + sbc/scaled_cell.R ... BENCH=1 NATIVE=1 with a throw-away scaffold card (§2.4)	compiled vs R path max

step	result
the RDM A = 0 card `rdm_ss_base_s801_e271` (28020b7): sha256 check, R port vs golden, `register_nn_model(card, twin = RDM)`	sha256 matches the Gadi bundles; R port vs golden 575 rows:
the LAN card registered with its twin (`sbc/lan_case5/lan_model.R`)	registers (type DDM, `c_name` NN). My hand comparison with the ORT golden rows returned the floor and I did not chase it; EMC2's own `test-nn-mlp.R` covers that comparison and passes

Not run here: anything on tux26, Gadi or AWS; training; the SBC cells; `nle_acceptance.R` (it loads the shipped artefacts by name and so hits the R 4.3.3 hash trap); `test-nn-validate.R` and `test-nn-reg-s402.R`.

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2. Worked example A: DDMnle (the scaled DDM)

2.1 How it was built with the current code

Sources: `handover/SCALED_DDM_PLAN.md`, `handover/nle-methods/simulation/scaled.R`, `simulation/sim_ddm_scaled.R`, `flow/scripts/train_ddm.py --scaled`, `retrain_classifier.py --scaled`, `export_flow_ddm.py --scaled`, `port/R/ddm_scaled.R`, `sbc/scaled_cell.R`, `flow/scripts/scaled_pipeline_tux.sh`.

step	what	where
parameterisation	the network sees (κ, η, Z, SZ) with κ = v a/s², η = sv a/s²; it models the unit-scale decision time u = (rt − t0)(s/a)² of a diffusion with a = s = 1; a and s are an exact rescaling, t0 an exact shift, st0 an exact difference of two CDFs: L = [G(u_1) − G(u_0)]/st0; exact zeros η = 0, SZ = 0; st0 = 0 is the plain density	`SCALED_DDM_PLAN.md:11-35, 62-75`; `ddm_scaled.R:1-16, 75-96`
encodings	κ identity; η and SZ `asinh(x/c)` with c = (0.01, 0.05), exact 0 at 0; Z probit	`scaled.R:13`, `ddm_scaled.R:25-40`; the card records `context_transforms = {kappa: identity, eta: zerobox, Z: probit, SZ: zerobox}` and `zerobox_c`
box and regimes	κ ∈ [−10, 10] (half the draws in [−3, 3]); η ∈ {0} ∪ [0.01, 8] (75 % log-uniform, 25 % asinh-uniform); Z ∈ [0.1, 0.9]; SZ ∈ {0} ∪ [0.05, 0.99]; regimes full 55 %, η = 0 15 %, SZ = 0 15 %, both 15 %	`scaled.R:11-25`
simulation	EMC2's `make_data` at a = s = 1, st0 = 0, fixed T0SIM = 0.05; the stored `ll` from `dDDM` at precision 5e-10; each chunk rewritten to columns `param_id, R, rt(=u), ll, kappa, eta, Z, SZ`; 600k draws × 40 rows (seed 5101), validation 25k × 40 (5102); Sobol with a seeded Cranley–Patterson shift	`scaled.R:27-57, 59-65`; `sim_ddm_scaled.R:1-9, 39-41`; 16 min on tux26 (`SCALED_DDM_PLAN.md:132`)
classifier target	exact P(R = 2	κ, η, Z, SZ) from `WienR::pWDM` at a = 1, sw = 2 SZ min(Z, 1 − Z), precision 1e-8
flow	conditional rational-quadratic spline over log u, [256, 128] conditioner, 64 bins, spline range [−12, 2] set from the pilot's data, batch 1024, 300 epochs, 30 checkpoints kept, 6 seeds × 4 threads = 24 cores, 26 h (07:29 14 Oct → 09:21 15 Oct)	`SCALED_DDM_PLAN.md:39-43, 179`; `scaled_pipeline_tux.sh:84-88`
classifier	`clf_sc_big` on the exact targets, mean	err
export and gate	float64 NumPy port of the network produces the golden vectors; gated against JAX **float64** at 1e-8 (`--ref f64`, the default under `--scaled`), because the float32 JAX reference itself differs from the	`export_flow_ddm.py:243-260, 425-435`; `SCALED_DDM_PLAN.md:159-181`; the diagnostic `flow/scripts/port_x64_check.py`

step	what	where
	port by up to 2.8e-2 in the far right tail; 180 cards exported in 5 min, worst 1.8e-10	
screening	score bias in 7 regimes, two stages (100 draws on all 180 cards, then 300 on the leaders); score = Σ over regimes of	g_v
SBC	18 cells (3 cards × 6 models) on AWS, N = 400, 500 replicates, network only, against the existing analytic controls; `sc_s1106_e245` outside the 95 % envelope once in 35 tests → adopted as DDMnle	`SCALED_DDM_PLAN.md:194-226`; `sbc/aws/scaled_sbc/run1/`
evaluation in EMC2, R path	`DDMsc(card)`: `DDM()` with `dfun` replaced, `c_name` dropped; transforms, bounds, zero exceptions and `rfun` are DDM's own	`ddm_scaled.R:98-109`
evaluation in EMC2, compiled path	the patch plus `ddm_sc_native_reg()`; agrees with the R reference to ≤ 1.1e-11 (laptop), 7e-13 here	`ddm_scaled.R:115-135`; `SCALED_DDM_PLAN.md:245-264`

2.2 What the package would need to reproduce it

A model specification (the shape is in §4.3) that says: inputs `kappa, eta, Z, SZ` with those transforms and constants; the box and the regime mixture above; a simulator that is a thin wrapper on EMC2's `make_data` plus the column rewrite; a choice classifier with an exact target function; `pre = "t0"` plus a scaling rule and a CDF-difference rule. The last two are the only genuinely new things in the whole example: the flow and classifier machinery is the same as every other DDM build. On the EMC2 side the card needs one more field than `flow_joint` has today (a `scaling` rule, now a free-text string at `export_flow_ddm.py:464-466`) and the loader needs to read it, which is what the patch does in C++ and nothing does in R.

Commands today: `Rscript simulation/sim_ddm_scaled.R train|val|choice`, `train_ddm.py --scaled ...`, `retrain_classifier.py --scaled --config big --out-json`, `export_flow_ddm.py --scaled --step N`, then `sbc/scaled_screen/run_stage{1,2}.sh`, `sbc/scaled_select.R`, `sbc/scaled_cell.R`.

2.3 Speed, as measured

`SCALED_DDM_PLAN.md:255-264` (laptop, N = 400): compiled DDMnle 0.075–0.125 ms per likelihood against WienR 0.33–1.22, 2.9x–9.8x; this machine 3.8x (full) and 1.6x (st0 only). The report (`REPORT_RUSS_JULIAN.md:22-35`) gives the per-model table. Against a continuous trial-level regressor the flow conditions once per unique row and loses its advantage (`EMC2_INTEGRATION.md:251-288`); nothing in this plan changes that, and the installation docs should say so.

2.4 What is still missing, and what I found

1. **ONNX export of DDMnle does not exist.** `export_onnx_ddm.py` builds the 8-parameter graph; DDMnle needs the network's CDF in the graph (spline CDF and a normal CDF through `Erf`), the preprocessing κ, η, c from the natural parameters, the t0 shift and the st0 branch, and an acceptance test against the compiled path to ~1e-11 (`REPORT_RUSS_JULIAN.md:152-167`). Probit and log stay with the caller.
2. **The committed DDMnle card has no golden vectors.** `port/export/` holds `ddm_sc_s1106_e245_weights.json` (and the two runners-up) but the only scaled golden file is `ddm_sc_pilot_e27_golden.csv`, whose weights are *not* committed; `port/tests/test_ddm_scaled_card.R` therefore cannot run on the committed material (it stops at `normalizePath`). **No tux26 run is needed to fix this** (probe, 2026-10-16, §13): the exporter's float64 NumPy port (`export_flow_ddm.py:172-204`, `ddm_eval_f64`) reads the weights JSON alone, so the golden grid (seed 20260708, 25 sets on the network-scale box with exact zeros forced, rt = geomspace(1e-4, 4, 23), both responses) can be regenerated from the committed card; EMC2's compiled evaluator reproduces that regenerated golden set to max |Δlog pdf| 1.24e-10 and |ΔCDF| 6.3e-15 over 1150 rows. The JAX float64 gate (which does need the checkpoint) was already passed at export, worst 1.8e-10 (`SCALED_DDM_PLAN.md:181`), so a golden file made this way is the

port's values gated once against JAX, which is what the pilot's golden file is too. **This is the first thing the verification command needs**, and it is a script, not a compute run.

3. **The compiled path is not reachable from `register_nn_model()`.** It needs the patch (uncommitted on the EMC2 side), `ddm_sc_native_reg()` (in the nle repo), and a zero-box 8-parameter scaffold card, which is not committed. For the probe in §1.3 I made a throw-away scaffold from the committed `ddm_cap256w_c4` card by relabelling `sv, SZ, st0` as `zerobox` with the zero-box constants and bounds; it is in this session's scratch space only and proves only that the plumbing works. The right fix is a card kind (say `flow_scaled`) that the R loader understands, so the scaffold disappears.
4. **Tail mass (corrected 2026-10-16, M0).** With `range_max = 2` on log u and identity tails, the spline hands over to the identity tail at $u = e^2$, and exactly $1 - \Phi(2) = 2.275$ % of the mass lies beyond that point for every θ (the sum of the two defective CDFs at `range_max` is 0.97725 for five random θ). That mass is **not lost**: the flow is defined on the whole positive axis and integrates to 1 over it; a quadrature that stops at `range_max` reads 0.9757 only because it stops there. So `nn_total_mass()` over its default grid (to 20 s) reads 1 up to the grid's own truncation, and the card states `support = (0, Inf)`, `mass_tolerance`, and in words how the tails carry the remainder (§13.3.4). The earlier wording of this item ("nn_total_mass will read 0.977") was wrong.
5. **The two score-bias regimes not covered by the committed material**: the screen scripts under `sbc/scaled_screen/` reference cards under `port/export/` on the laptop; only the three selected cards are committed, so the selection cannot be re-run from the repository (by design: `RECIPE.md:17-18`, regenerate rather than deposit).

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3. Worked example B: the RDM

The race model is the generality case, not the speed case (`METHODS.md:73-78, 106`): the flow models one accumulator's first-passage density f and survivor S, and the race likelihood is assembled as $L(rt, R \mid \theta) = f_R(rt) \cdot \Pi_{j \neq R} S_j(rt)$ (`METHODS.md:127-135`; `port/R/flow_race.R:9-17`). No classifier.

3.1 The simulator, box and the three RDM flows that exist

	`rdm_small` (committed)	**A = 0 flow `rdm_ss_base_s801_e271`**	**Malte's RDM/CRDM (converted)**
context	`v, B, t0, s, A`, all `log`, standardised by the training moments	same, with `A` as `zerobox`, c_A = 0.05 (`RDM_A0_BOX_PLAN.md:25-30`)	`v, s, b` (RDM) or `v, amp, tau, s, b` (CRDM), **natural scale**, `input_scaling` = (log(x + ε) − loc)/scale, no `scaler` (`MALTE_FLOWS_PROMPT.md:33-45`)
box (natural)	lower (1e-3, 0.1, 0.05, 0.1, 1e-4), upper (5, 3, 1, 2, 2) (`export_flow.py:74-94`)	A ∈ [0, 2], continuous to 0; three-part mixture 70 % uniform / 15 % asinh-uniform / 15 % exactly 0 (`RDM_A0_BOX_PLAN.md:32-46`)	v [0, 8], s, b [0.25, 3.5] (CRDM: amp [0, 1], tau [0.01, 0.5], s, b [0.25, 3])
data	16e6 rows, one trial per draw, analytic `ll` stored, seed 3876 (`simulation/sim_rdm.R`)	24e6 rows, seed 4876, simulated on tux26 in under a minute (`RDM_A0_BOX_PLAN.md:161-167`)	online, 64 θ × 500 trials per step, nothing stored (`MALTE_CRDM_PLAN.md:36`)
architecture	[64, 32], 64 bins, range ±5 (`METHODS.md:138-143`; card `rdm_rdm_small_e195_weights.json`)	[256, 128], 64 bins (`RDM_A0_BOX_PLAN.md:50-52`)	[128, 128], **12 bins**, per-context affine loc/scale, `pre = "t0"` (decision time)
selection	checkpoint e195 of 30, by score bias (`METHODS.md:219-220`)	e271 of the seed study's baseline arm (120 screened checkpoints,	single run, no selection

	rdm_small (committed)	A = 0 flow rdm_ss_base_s801_e271	Malte's RDM/CRDM (converted)
		MALTE_STAGES.md "M4 part E")	
SBC	N = 100 and 400, 500 reps, all parameters inside the gate, ranks uniform (METHODS.md:224-233)	Gadi, 2026-10-06, final sampler: A = 0 cell N 400 (250 reps) and standard cells N 100/400 (500 reps), network and control,	std bias
weights in the repo	**yes**: handover/nle-methods/artefacts/rdm_rdm_small_e195_weights.json and rdm_small.rds; golden rdm_rdm_small_e195_golden.csv in the EMC2 test tree	**no** (the Gadi bundles hold cell.rds, run_cell.R, session.txt; the card lived in port/export, untracked)	**no** trained weights; three random-weight fixtures with golden vectors under malte/fixtures/

Decided (AH, 2026-10-16, §13): the worked example is the A = 0 box flow, checkpoint rdm_ss_base_s801_e271, selected from 120 checkpoints by W = max |g| (W = 0.0022; the flow of the report's Appendix C), not Malte's and not rdm_small. Its card, golden vectors and intermediates are on this branch since 28020b7: handover/nle-methods/port/export/rdm_ss_base_s801_e271_{weights.json,golden.csv,intermediates.json}. The weights file's sha256 is b6ce7db9ca98ed17f1b04ecfc8860744fd5716ee20709f1baddcccdac0b44700, verified here against the hash the Gadi bundles record (sbc/gadi/cells/rdm_o_std_N400/bundle/session.txt:2). The golden CSV (575 rows, 25 parameter sets × 23 rt, five sets at exactly A = 0) is the RDM reference for every port test, as the existing exports are for DDMnle; nothing is regenerated from checkpoints. The final release location is decided at packaging time.

3.2 Screening, the seed study, and the EMC2 side

Score bias for a race model: handover/nle-methods/evaluation/score_bias_race.R (H = 0.01, the trapezoid measure, t0 by the restricted-region estimator; analytic control ~1e-4, METHODS.md:157-182). The seed study (handover/MALTE_STAGES.md M7a/M7b, five arms × four seeds: baseline, affine, decision time, both, Malte's recipe; sbc/seed_study_launch_tux.sh, sbc/seed_study_screen.sh) has its baseline arm screened (sbc/seed_study/ss/summary_base.csv); the full screen and its reading are the handed-over stage at the bottom of MALTE_STAGES.md and are not in the log as of this commit. So whether --affine and --decision-time are part of "the recipe" or experiments is still open, and the package should carry them as options, not defaults.

EMC2 side: register_nn_model(card, twin = RDM); for a card without A, pars = c("v","s","B") with exceptions = c(A = 0) (sbc/gadi/cells/build_rdm_cell.R:30-45); Malte's cards register with hybrid = "amp" so that the unpulsed accumulator is scored by the closed-form Wald (nn_register.R:665-682). The compiled race evaluator takes the input_scaling and affine fields (src/flow_race.cpp:153-200).

3.3 What the package would need to reproduce it

A spec with: context v, B, t0, s, A (or v, s, b with A fixed at 0), transforms log/zerobox, the box, the A mixture; a single-accumulator simulator wrapper on EMC2's rWald writing rt, ll, param_id and the context; response = "race" (no classifier; f and S from one network); optional pre = "t0" and affine. The Malte path is a converter (flow/scripts/malte_to_card.py) rather than training: it reads an Orbax checkpoint plus a sidecar and writes a card with context_scale = "natural", input_scaling, affine, affine_scale, pre (:124-151).

---

4. What the two share, where they differ, and the interface that follows

4.1 Shared (the package core)

- A conditional rational-quadratic spline flow over a log time, one spline, standard-normal base, identity tails, GELU-tanh MLP conditioner; the float64 NumPy port that generates golden vectors; the port-vs-

JAX gate.

- The simulate → train (one run, large batch, many checkpoints) → export → screen on score bias → select → SBC-with-matched-control recipe (`RECIPE.md`).
- The card: `context_names`, `context_transforms` (+ `zerobox_c`), `bounds_natural`/`bounds_sampled`, `scaler` or `input_scaling`, `mlp`/`flow_mlp`, `spline`, `ll_floor_log`, `pre`.
- The validation kit in EMC2, which is model-agnostic given a registered model with an analytic twin.

4.2 Different (the model-specification interface)

axis	DDMnle	RDM	what the spec must carry
response structure	two responses; P(R	θ) from a separate classifier on an exact target; the flow sees R as an input	race; one network per accumulator, f and S; no classifier
exact zeros	η = 0, SZ = 0 (asinh encoding); st0 = 0 is the plain density	A = 0 (asinh); Malte: amp = 0 is the Wald (hybrid)	per-input `zerobox` with its constant; stratified regimes
closed-form transforms	a, s rescaling; t0 shift; st0 by CDF difference	t0 shift (decision time)	`pre` (shift) and a `scaling` rule (new: §7.1)
analytic control	yes (WienR / `dDDM`)	yes (`dRDM`, Wald); CRDM: Volterra reference	the twin's name, or a reference `dfun`
input scale	sampled (EMC2) with a `scaler`	sampled, or natural with `input_scaling` (Malte)	both; the loader has both
spline range	[−12, 2] (set from pilot data)	[−5, 5]	a spec field, not a constant

4.3 The hard-coded pieces this exposes

Python (`handover/nle-methods/flow/`): `CONTEXT*` literals (`train_ddm.py:43-62`, `export_flow_ddm.py:45-61`, `retrain_classifier.py:31-56`), `CONTEXT`/`MODEL_REGISTRY` (`train_race.py:71-74`, `export_flow.py:74-94`); `TRANSFORMS`/`BOUNDS_NAT`/`ZEROBOX_C`/`ZEROBOX_BOUNDS_NAT` (`export_flow_ddm.py:74-75, 111-119`; `data.py:87`; `train_race.py:102`; `export_flow.py:69`); the spline range fixed at ±5 in the race exporter (`export_flow.py:372-375`) but read from `variant_config.json` by the DDM one; column names `rt, R, ll, param_id, p_R2, d`; binary R with the P(R = 2) logit (`config.py:269-276`); the edge loss's DDM column indices (`training.py:65-66`); `SHIFT = "t0"` (`train_race.py:79`) and `pre = "t0"` always in `malte_to_card.py:129`; `results/model_states/<tag>_budget/{DDM_flow, DDM_classifier,RDM,LNR}` and `port/export/` paths; `REPO_ROOT = parents[2]`; the classifier placeholder copied from `--classifier-from large` (`train_ddm.py:219-230`); golden seeds and rt grids.

R: `DDM_SC_INPUTS`, `DDM_SC_C` and the eight-column `sc_pars` (`ddm_scaled.R:25-26, 133`); `SC_*` constants and regimes (`scaled.R:13-25`); `scaled_cell.R`'s scaffold and regime table (`:27-31`); the eight-case drivers.

EMC2: joint kinds are two-response (`model_NN.h:165-169`, `nn_register.R:1108-1109`); twin table DDM, RDM, LNR, LBA, CRDM (`nn_register.R:59`); zero-box names `sv, SZ, st0` and the `st0zero` rule (`:601-619`); the support rule looks up `st0` and `t0` by name (`flow_ddm.cpp:281-298`); hybrid Wald map `c(v, b → B, s)` (`:674`); the `triple` encoding's column positions (`flow_ddm.cpp:68-80`); the scaled patch's exactly-four-inputs and eight DDM columns.

4.4 Details the documentation needs that are not written down (answered by AH 2026-10-16; see §13)

1. Which RDM artefact is the worked example (§3.1), and whether the A = 0 flow's card should be committed.
2. The SZ convention at every interface: the network's SZ is the *relative* width (SZ/m, `ddm_scaled.R:30-35`); EMC2's `DDM()` passes `sw = 2 SZ min(Z, 1 − Z)` after `Ttransform`; the registered scaffold passes a proportion (`scaled_cell.R:53-57`). This is in comments only.
3. The s = 1 rule in every RDM control cell (`build_rdm_cell.R:12-13`, "pWald overflows for v b/s² > 354") and the ±4 sd prior-inside-box rule (`nn_validate.R:94`, `sbc_specs.R:6-9`): are these package defaults or cell conventions?
4. Whether `--affine` and `--decision-time` are recipe or experiment (the seed-study reading is pending).

5. The tail-mass convention (§2.4.4) and what `nn_total_mass()` should expect for a card.
6. The t0 correction's status for the old DDM artefacts (superseded by DDMnle for case 8; `OPEN_ITEMS.md:31-35`): the package should not carry it, but the documentation of `cap256w_c4` must say what applies.
7. `rt_resolution ≤ 1/1000` for t0-sensitive work (`METHODS.md:714-715`): a package default or a vignette note?
8. Where the DDMnle checkpoints are (tux26, `~/emc2nn/nle/handover/nle-methods/results/model_states/sc_s1106_budget` by the pipeline's layout) so the golden vectors can be regenerated (§2.4.2).

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5. Scope and audience

who	what they cannot do now	what they would do with this	needs
us (Julian, Malte, AH)	reproduce DDMnle or the RDM flow without the laptop's `port/export`, tux26 checkpoints and a session's memory of which flag; verify a card without the nle repo	re-run either example from a spec on a fresh machine; add the next model (CRDM-style, LBA variants) by writing a spec; ship cards with golden vectors	the Python package + the EMC2 pieces in §0
EMC2 users with an intractable model	train anything; the only route is to copy `flow/` and edit six files; nothing documents the traps in one place	write a simulator and a spec, run one command on a workstation or cluster, get a card, register it, run the kit before trusting it	the same, plus the installation docs of §9
HSSM / LANfactory users wanting EMC2	nothing they cannot do today in principle: `onnx_to_card.py` exists and the LAN registers with a hand-written twin (`sbc/lan_case5/lan_model.R`); but it supports tanh MLPs with a rank-1 input only, one preset, and the user must know the parameter mapping and must run an SBC cell before inference (`EMC2_IMPLEMENTATION_BRIEF.md:175-184`; the one LAN scored carries g_a = −0.122, `METHODS.md:434`)	a converter that reads the HSSM model config for the mapping and box, accepts batched graphs and ReLU/other activations, and refuses what it cannot represent	the converter moved into the Python package and extended; a twin-writing helper in EMC2
HSSM users wanting DDMnle	nothing: the DDMnle ONNX graph does not exist (§2.4.1)	load one ONNX file with the LAN-shaped signature	the ONNX exporter for the scaled kind, and an HSSM model config (outside our control)

Overlap. EMC2 `dev-nle` already does the import, validation and fitting half; this plan does not move any of it out. HSSM/LANfactory train LANs (per-trial MLPs, Python, ONNX out) and have nothing for flows with amortised conditioning or for calibration beyond parameter recovery; our converter consumes their output. BayesFlow and sbi do amortised posterior/likelihood estimation in Python with their own samplers, not EMC2's; the only overlap is the word "flow". Nothing in the field ships the score-bias/posterior-shift/SBC-with-control kit, which is EMC2's.

Where each piece lives

piece	lives in	why
model spec, simulator glue (parquet writer), training, checkpoint export with golden vectors, float64 gate	Python `nleflow`	JAX; the port that makes the golden vectors is NumPy
ONNX export (both signatures, all kinds), `onnx_check`	Python `nleflow`	onnx helper API
ONNX → card converter	Python `nleflow` (moved from `EMC2/inst/scripts`)	it already requires Python; EMC2 keeps a one-line pointer

piece	lives in	why
card format spec and version, loader, `register_nn_model`, compiled evaluators, the kit, `nn_golden_check`	EMC2	they are there, tested, and must not be duplicated
R reference evaluators (`flow_race.R`, `flow_ddm.R`, `ddm_scaled.R`)	nle repo now; EMC2 `tests/testthat` fixtures (two of three already are, `test-nn-port.R`)	the readable spec of the C++
the two worked examples (specs, small smoke data, cards, golden, SBC results)	`nleflow` examples + the nle repo's `handover/nle-methods/`	vignettes and end-to-end tests
Malte converter (`malte_to_card.py`)	Python `nleflow` as an optional "converters" module	its environment (jax ≥ 0.9) conflicts with ours; keep it isolated

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6. Architecture

option	what it is	for	against
A. R package wrapping Python (reticulate)	`nleFlow::train(spec)` calls JAX through reticulate; export, convert and validate in R	one language for the user; `design()`-style spec in R	reticulate + a pinned JAX env + optional CUDA in an R package: not CRAN-able, fragile across JAX releases (0.5.3 here, 0.10.2 for Malte, both in this repo); every evaluator would be re-implemented in R or copied from EMC2 (drift); the dedicated-library trap gains a Python twin
B. Python package + R front end	`nleflow` (Python) does spec, train, export, ONNX, convert; EMC2 does cards and validation; a thin R helper (in EMC2 or `nleflow`'s `R/` dir) writes the spec JSON and the parquet from `make_data`	each half in its natural language; EMC2 stays the single R evaluator; the R helper is ~200 lines	two installs (R + Python); Python needed to train, which is true of every option
C. R-only tools + documented recipe	an R package for export/convert/validate; training stays "run these scripts"	smallest	duplicates `dev-nle`; converting ONNX needs Python anyway; the scripts stay unusable for a new model

Recommendation: B, with the R front end inside EMC2 rather than a separate package. The R parts that must be R are: the simulator glue (`make_data` → parquet; this is the one place the user's EMC2 model function is needed), the card loader and compiled evaluators, the kit, and the golden check. All of those exist in EMC2 except the parquet writer (today `simulation/sim_functions.R`'s `simulate_data` in the nle repo, with `arrow`) and the golden check.

Dependency and installation burden. EMC2 `dev-nle` adds `jsonlite` (Suggests, `DESCRIPTION:47`) and `loo` (Imports) over the 3.5.0 release, and pins `WienR (>= 0.3-17)`; no reticulate, no onnx. It needs a C++17 compiler, as EMC2 always has. The Python side is a pinned `uv` environment (11 s here). GPU is optional (`jax[cuda12]` extra); DDMnle trained on CPU. CRAN: the EMC2 NN code is CRAN-compatible as far as dependencies go; whether `dev-nle` goes into the next release is AH's and Niek's call and outside this plan. The Python package goes to PyPI or stays a GitHub install; either way with a lock.

The dedicated-library trap stays whatever is built, because `dev-nle` and the release share a version prefix (3.4.1.9001 vs 3.4.1 in the dev-nn era; the ampl-psych checkout here is 3.5.0). The docs and the verification command check `find.package("EMC2")`, never the version (§9.3).

---

7. Interfaces

7.1 The card format: version it, write it down

Today the card is whatever `export_flow*.py` writes and `nn_normalise_card()` / the C++ loaders accept (`nn_register.R:155-180`; `flow_ddm.cpp:256-309`; `flow_race.cpp:153-200`; `mlp_lik.cpp:90-141`; `nle_flow.h:66-97, 170-181`). There is no version field and no document; `man/register_nn_model.Rd` describes the fields in prose. Proposed `card_version = 1`, with the spec living in EMC2 (`vignettes/` or `inst/doc`) and mirrored in `nleflow`:

group	fields	read by
identity	`card_version`, `kind` (flow_joint / flow_race / mlp_joint / regression_joint / **flow_scaled**), `model`, `label`, `source`, `sha256` of the weights	R
inputs	`context_names`, `context_transforms` (identity, log, probit, zerobox), `zerobox_c`, `context_scale` (sampled/natural), `scaler` or `input_scaling`, `bounds_natural`, `bounds_sampled`	R and C++
structure	`pre` (shift parameter), `hybrid`, `response_coding` / `response_values`, `input_layout` (MLP kinds), **`scaling`** (the DDMnle rule, as a structured field: `{u: "(rt - t0) * (s/a)^2", inputs: {kappa: "v*a/s^2", eta: "sv*a/s^2"}, smear: "st0"}` or an enumerated kind)	R and C++
network	`mlp/flow_mlp/classifier_mlp` (`activation`, `use_norm`, `layers`, `norms`), `affine`, `affine_scale`	C++
spline	`num_bins`, `num_splines = 1`, `range_min`, `range_max`, `min_bin_size`, `min_knot_slope`, `boundary_slopes = identity`, `output_transform = exp`, `base_distribution`	C++
floors and mass	`ll_floor_log`, **`tail_mass`** ($1 - \Phi(\text{range_max}) + \Phi(\text{range_min})$, §2.4.4)	R
provenance	`budget`, `checkpoint_step*`, `exported`, `exporter_version`, `golden` (path or embedded hash of the golden file), `validation` (score-bias, posterior-shift and SBC records with dates)	R, not required

Rules: unknown fields are carried along; a card without `card_version` is version 0 and loads as today; a loader refuses a version it does not know. `register_nn_model()` already refuses mis-ordered contexts and missing transforms; it should additionally refuse a card whose golden file is present and fails (`nn_golden_check`).

7.2 The model specification a user supplies (Python; an R helper writes the same JSON)

```
        model: RDM_A0                # a name; also the twin's name when it is an EMC2 model
        response: race               # race | choice
        context: [v, B, t0, s, A]    # in network order; asserted everywhere
        transforms: {v: log, B: log, t0: log, s: log, A: zerobox}
        zerobox_c: {A: 0.05}
        box_natural: {v: [1e-3, 5], B: [0.1, 3], t0: [0.05, 1], s: [0.1, 2], A: [0, 2]}
        sampling: {A: {uniform: 0.70, asinh_uniform: 0.15, zero: 0.15}}   # strata; the rest uniform in the
transform
        pre: t0                      # optional: the flow models rt - t0
        scaling: null                # DDMnle: {kind: ddm_unit}
        spline: {bins: 64, range: [-5, 5]}
```

```
        conditioner: [256, 128]
        classifier: null              # choice models: {target: p_R2, widths: [256, 256, 256]}
        training: {rows: 24e6, trials_per_draw: 1, batch: 1024, epochs: 300, keep: 30, peak_lr: 1e-4, seeds:
[801, 802, 803, 804]}
        simulator: Rscript simulate.R  # anything that writes the parquet columns below
```

Parquet contract (today's loader, `data.py:107-160`): one row per trial, columns `param_id`, the context on the sampled scale (−Inf at exact zeros), `rt` (or the pre-shifted time), `R` for choice models, `ll` (the analytic log density where it exists; the loader reads it even for plain MLE, so make it optional in the package), and for choice models a second file with the context plus `p_R2`.

7.3 ONNX signatures

Two, as now (`export_onnx_ddm.py:9-27, 92-107`): **trial-wise** `[N, k+2]` (context on the sampled scale, rt, choice ∈ {−1, +1}) → `[N]` log p, the LAN-shaped contract for HSSM; and **shared-theta** `theta[k], rt[N], choice[N]`, 25x faster because the conditioner runs twice and the classifier once (`EMC2_INTEGRATION.md:164-179`). Both float64 (float32 rejected at 1.9e-4, `:185-186`); opset 18, IR 10 for onnxruntime ≤ 1.30 (`:264-266`; 1.31.0 here loads them). For `flow_scaled` the graph takes the natural DDM parameters and does κ, η, c, the shift and the CDF difference inside (`REPORT_RUSS_JULIAN.md:160-165`), with probit and log left to the caller. For race cards an ONNX export is possible per accumulator (f and S), but no consumer exists; defer.

7.4 The ONNX → card converter

Supported today (`onnx_to_card.py:69-147`): one non-initializer input, `MatMul`+`Add` or `Gemm`, `Tanh`, `Identity`, one linear output; rank-1 input assumed by `--verify` (`:263-266`); one preset. To add, in order of value: batched `[N, k]` inputs (HSSM batches by converting the graph; `EMC2_IMPLEMENTATION_BRIEF.md:123-131`), `Relu`/`Gelu` activations (the evaluator `load_mlp` accepts `gelu_tanh` and `tanh` only, `nle_flow.h:66-97`; each new activation is a C++ change too), a preset per published LAN with its parameter mapping and box read from the HSSM model config, and a refusal message that names the first unsupported node. The converter must keep writing `calibration: "unknown; needs its own SBC cell before inference"` (as the LAN card does: `sbc/lan_case5/lan_ddm_uniform_st.rds`).

7.5 Validation built into export, so that a card cannot be exported without passing

gate	where it runs	exists
port vs JAX float64 ≤ 1e-8 (float32 gap reported)	exporter	yes for the DDM (`export_flow_ddm.py:429-435`); the race exporter gates against float32 only (`export_flow.py:471-507`)
golden vectors written with the card; `nn_golden_check(card)` in EMC2 reproduces them to 1e-10	exporter + EMC2	golden yes; the R check only as test code
context order and transforms asserted on load	EMC2	yes
total mass = 1 − tail mass	EMC2 kit	`nn_total_mass` exists; the expectation needs the card field
score bias with its analytic control, control < 1e-3	EMC2 kit / `score_bias_*.R`	yes
posterior-shift screen	EMC2 kit	yes
one SBC cell with a matched control; result recorded in the card's `validation` block	EMC2 kit	the cell yes; the record no

The export command should refuse to write a card whose port gate fails, and `register_nn_model()` should warn when a card has no `validation.sbc` entry (that is the honest default: "loads, not yet fit to infer with").

---

8. Reusable now, needs generalising, missing

Reusable as is (would be moved, not rewritten): the NumPy float64 port and gate (`export_flow.py:116-254`, `export_flow_ddm.py:172-204, 380-451`); `export_onnx_ddm.py` (graph construction); `onnx_check.py`; `onnx_to_card.py`; `malte_to_card.py`; `port_x64_check.py`; the loader `PreloadedBatches` and the training loop (`training.py:470-596`), minus the DDM-specific losses; `checkpoints.py`; the R reference evaluators (`port/R/*.R`); the score-bias and posterior-shift estimators (already in EMC2 as `nn_score_bias`, `nn_posterior_shift`); `simulate_data` (`simulation/sim_functions.R`) as the parquet writer; EMC2's whole NN layer.

Needs generalising: everything in §4.3. Concretely: one `train` command that takes a spec (replacing `train_ddm.py`/`train_race.py`); one `export` command (replacing `export_flow.py`/`export_flow_ddm.py`), with the float64 gate for every kind and the spline range from the spec; the classifier step without the copy-a-placeholder-checkpoint dance (`train_ddm.py:219-230`); `apply_context_encoding` keyed by the spec's transforms rather than by parameter name (`data.py:90-105`); the race exporter's ±5 refusal; `malte_to_card.py`'s `pre = "t0"` always; EMC2's `flow_scaled` kind in R; EMC2's two-response assumption left as a documented limit for now.

Missing: the DDMnle ONNX graph (§2.4.1); DDMnle golden vectors for the committed card (§2.4.2); `nn_golden_check`; the card version and spec; `tail_mass`; a verification command (§9.4); the R < 4.5 hash fix (`nle_file_hash` writes md5 into `MANIFEST` or refuses R < 4.5 with a clear message); `onnx`/`onnxruntime` in the Python lock; a README and `requirements.txt` in the handover `flow/` (`pyproject.toml` declares `readme = "README.md"`, which is absent there, so an editable install of the handover copy would fail); the training loop's `avg_loss` bug for sets under 10 batches (`training.py:432-436, 463-467`); the end-to-end smoke tests of the two examples on 1 % data (the pattern exists: `sbc/seed_study/a0smoke/`, `m7asmoke/`, the 2-epoch smoke in `SCALED_DDM_PLAN.md:115-122`).

---

9. Installation documentation (scoped; what I tested)

9.1 Inventory per function group

function group	needs	version actually used (evidence)	not recorded
use a card in EMC2 (fit, kit)	R; EMC2 `dev-nle` built from source (C++17 compiler, OpenMP optional); Rcpp, RcppArmadillo, WienR ≥ 0.3-17, jsonlite, and EMC2's Imports	laptop R 4.6.0, macOS 14.5, reference BLAS, WienR 0.3-17, Rcpp 1.1.2, jsonlite 2.0.0 (`sbc/gadi/cells/*/bundle/session.txt`); Gadi `module load R/4.5.0`, gcc 12.2.0 + intel-compiler 2021.10.0, MKL, `-fp-model=precise` (`sbc/nci/*.pbs:23-25`, `MALTE_STAGES.md:1110`); here R 4.3.3	tux26 and AWS: no R version anywhere; macOS compiler/Xcode; **R < 4.5 breaks shipped artefacts** (§1.3)
train	Python 3.11–3.12, `uv`, the lock (jax 0.5.3 etc.); `arrow` in R for the simulator (plus `randtoolbox` for Sobol in the scaled simulator); EMC2 for `make_data`	tux26 python 3.12 (`MALTE_STAGES.md:338`); laptop interpreter version unknown; CPU only for every run that shipped	GPU never used; the laptop's Python patch version
export to EMC2 card	the same Python env (NumPy port)	as above	
export to ONNX	the Python env + `onnx` (+ `onnxruntime` to check)	"onnxruntime 1.30, onnx 1.23" in a comment only (`export_onnx_ddm.py:264`); here onnx 1.23.2, ort 1.31.0	which machine; not in any lock
convert ONNX → card	Python + `onnx` + numpy (+ onnxruntime for `--verify`)	as above	
SBC and score-bias checks	EMC2 `dev-nle`, WienR; cores	laptop 8 cores (policy); AWS c7a.32xlarge 128 cores, 6.45 h, ≈ $45 for 18 cells (`SCALED_DDM_PLAN.md:196-208`); Gadi 48 cpus per cell job	
Malte's converter	its own env: Python 3.11, jax 0.10.2, flax 0.12.8,	recorded	

function group	needs	version actually used (evidence)	not recorded
	malte/requirements.lock		

9.2 Per-platform steps (each with its one-line check)

Order matters: R and its packages, then EMC2 into a dedicated library, then the card check, then Python only if training or ONNX is needed.

macOS (Apple Silicon; what the project ran on) 1. R ≥ 4.5 from CRAN's installer. Check: `R --version | head -1` shows ≥ 4.5 (4.3/4.4 hit the hash trap). 2. Xcode command-line tools and Homebrew `libomp` (EMC2's `configure:104-151` links Accelerate and uses libomp for OpenMP). Check: `clang++ --version` prints; `R CMD config CXX17` prints a compiler. 3. `install.packages(c("Rcpp","RcppArmadillo","WienR","jsonlite","arrow","randtoolbox", <EMC2 Imports>))`. Check: `Rscript -e 'stopifnot(packageVersion("WienR") >= "0.3.17")'`. 4. EMC2 `dev-nle`: `R CMD INSTALL --preclean --library=~/R/nle-lib <EMC2 source>` (the tarball, or `git clone -b dev-nle`). Check: `Rscript -e '.libPaths(c("~/R/nle-lib", .libPaths())); library(EMC2); cat(find.package("EMC2"), "\n")'` prints a path under `~/R/nle-lib`. 5. The verification command (§9.4). 6. (training/ONNX) `uv venv --python 3.11 && uv sync --frozen` in the Python package; `uv pip install onnx onnxruntime` until they are in the lock. Check: `python -c "import jax, flax, onnx, onnxruntime; print(jax.__version__)"` prints 0.5.3. Set `JAX_ENABLE_X64=1` only for the export gate (the exporter sets it itself, `export_flow_ddm.py:259-260`); training runs float32.

Not tested on macOS in this stage (no macOS here); the steps are what the repo did, with the checks added.

Linux workstation (tux26-like) 1. R ≥ 4.5 (CRAN's Ubuntu repository or a module). Check as above. On a shared machine with another EMC2 in the default library, `R_PROFILE_USER=/dev/null` for every run (as `sbc/aws/scaled_sbc/bundle/smoke.sh` does). 2. `g++` ≥ 11 (`CXX_STD = CXX17`), `libopenblas` optional. Check: `R CMD config CXX17`. 3. R packages as on macOS (Ubuntu: `apt-get install r-cran-{rcpp,rcpparmadillo,...}` is faster; WienR is not packaged). Check as above. 4. EMC2 `dev-nle` into `~/R/nle-lib` with `MAKEFLAGS=-j$(nproc)`. Check: `find.package`. 5. Verification command. 6. Python by `uv` as above; `OMP_NUM_THREADS` per run (the pipeline used 4 per seed, `scaled_pipeline_tux.sh:7`).

Cloud instance (this one; AWS recorded in `sbc/aws/scaled_sbc/bundle/setup.sh`) 1. `apt-get install r-base-core r-base-dev` (45 s here; R 4.3.3 on Ubuntu 24.04, so step 4's check matters and the shipped-artefact trap applies: use JSON cards, or R ≥ 4.5 from CRAN's repository). 2. `apt-get install r-cran-<each>` for the Imports (2 min 16 s), WienR from source (18 s), `loo`, `msm` as needed. Check: `Rscript -e 'for (p in c("WienR","jsonlite","Rcpp","RcppArmadillo","loo")) stopifnot(requireNamespace(p))'`. 3. `tar -xzf emc2_142bd497.tar.gz; MAKEFLAGS=-j$(nproc) R CMD INSTALL --preclean --library=/opt/emc2/lib <src>` (78 s here with 4 cores; `setup.sh:11-19` writes `EMC2_SHA.txt` beside the library, which is a good convention). Check: `find.package`. 4. Verification command. 5. `uv` + the lock (11 s here). Check as above.

Traps to print in the docs (all hit in this project): the dual install and `find.package` (`EMC2_INTEGRATION.md:8-24`); `R_PROFILE_USER=/dev/null` on shared machines; a card's absolute path is baked into a saved registration/bundle (`nn_register.R:709`, `nn_validate.R:1014-1015`; the Gadi bundles once "never fitted a replicate" for this, `MALTE_STAGES.md:1112-1114`); the float64 export gate for scaled cards; EMC2 ships its own compiled copy of the port, so edits to `port/src` prove nothing about EMC2; `prior()` positional means; `mapped_pars()` asserted for every parameter not sampled; R < 4.5 hash trap; `onnx`/`onnxruntime` not in the lock.

9.3 The single verification command

Proposed `nleflow-verify` (an R script shipped with EMC2 as `inst/scripts/nle_verify.R`, or `Rscript -e 'EMC2::nn_verify()'`), installing nothing. It: 1. prints `R.version.string`, `find.package("EMC2")`, `packageVersion("EMC2")` and the library's `EMC2_SHA.txt`, and stops if R < 4.5 with the hash-trap message; 2. loads each committed card (DDMnle `ddm_sc_s1106_e245`, `rdm_small`, `ddm_cap256w_c4`, the LAN card) through `register_nn_model()`, asserting context names and transforms; 3. evaluates the compiled likelihood at the golden rows and compares with the golden file to 1e-10 (`nn_golden_check`); for the LAN, with the onnxruntime golden CSV at 1e-4 (float32 source); 4. for DDMnle, also runs the oracle test (`port/tests/test_ddm_scaled.R`, 7 s) and the compiled-vs-R agreement on

200 draws (`scaled_cell.R`'s check, 7e-13); 5. prints one line per card, `PASS max|diff| 6.9e-13`, and exits non-zero on any failure.

What it can check today: `rdm_small` and `cap256w_c4` (golden in EMC2's test tree); the oracle test; the LAN card. What it cannot until §2.4.2 is done: DDMnle against golden vectors (none exist for the committed card), and the compiled DDMnle path through `register_nn_model()` (needs the `flow_scaled` kind, or the scaffold).

9.4 Scripted or documented?

Documented steps with one check per step, plus a container, plus the verification command; not a setup function in the package. A setup function that installs R packages and builds EMC2 into a chosen library is thirty lines and I would not trust it on a machine with another EMC2 on the path; `sbc/aws/scaled_sbc/bundle/setup.sh` is already that script for Ubuntu and should be kept as the cloud recipe (it is 20 lines and worked). A `Dockerfile` (Ubuntu 24.04, R from CRAN's repository so that R ≥ 4.5, the apt packages, WienR, EMC2 from the tarball or the branch, `uv` with the lock) reproduces what I ran here in about 5 minutes and gives a cluster a Singularity image. Effort: 2 days for the docs with checks, 1 day for the container, 2 days for the verification command (half of it the golden check in EMC2). Risk: low; the one real risk is the R-version trap, which the command detects.

9.5 What I ran and what I only wrote

Ran (this instance): §1.3's table in full; the cloud steps 1–5 above except the verification command, which does not exist yet (its pieces ran separately: oracle test, ONNX check, converter, EMC2's nn tests, the compiled-path probe). Wrote only: the macOS and Linux step lists (they are the project's recorded practice plus checks), the container, the verification command.

---

10. Effort, risk, maintenance

10.1 Work breakdown (days, one person who knows the code; Claude-assisted as the project has been)

item	days	who
M0 first milestone (§11.1): card spec v1 + version field; `flow_scaled` kind in R (wire the patch); `nn_golden_check`; DDMnle golden vectors regenerated (tux26 run, minutes); R < 4.5 fix; `nle_verify`	5–7	EMC2 session (Fable/Opus: trap-dense) + one tux26 export run
Python package skeleton: spec schema, loader keyed by spec, one `train`, one `export` with the float64 gate for all kinds, lock with onnx/onnxruntime, README	8–10	Sonnet-grade against the two examples' smoke tests
the two examples as specs + 1 % smoke tests + vignettes (DDMnle, RDM A = 0 or `rdm_small`)	3–4	
ONNX: generic exporter over card kinds; the DDMnle graph with CDF, preprocessing and st0 branch; acceptance vs compiled path on 200 draws × 6 models	4–5	needs the compiled path (M0)
converter: move into the package; batched inputs; activations; presets from HSSM configs; refusal messages	2–3	each new activation is also a C++ line in `load_mlp`
EMC2 rebase of `dev-nle` onto `dev` (the new sampler is already on both sides, differently) and re-running the usability gate	2–3	EMC2 session

item	days	who
(EMC2_IMPLEMENTATION_BRIEF.md:301-351, 25 min at 8 cores)		
installation docs, container, CI (GitHub Actions on Ubuntu: build EMC2 from the branch, run nle_verify)	3–4	
total	**27–36**	

Compute: no training is needed for any of it except the smoke tests (minutes). The DDMnle ONNX acceptance test is seconds. Regenerating golden vectors is one export on tux26 (ask first, as always).

10.2 Risks

risk	where it bites	mitigation
format drift between the exporter and EMC2's loader	the only interface that matters; today there is no version	version the card; a loader that refuses unknown versions; nle_verify in CI on both repos; EMC2's test-nn-* already pin the three shipped cards
EMC2 dev-nle vs dev	dev-nle is behind dev by the sampler commits of 6–9 Oct and ahead by the NN layer; 142bd497 and the scaled patch are local	rebase before M0; push the patch as a branch so it is not a file in another repo
ONNX operator coverage	the DDMnle graph needs Erf (opset ≥ 9, fine) and the CDF-difference branch; Where-heavy graphs are slow in onnxruntime for the trial-wise signature	keep both signatures; accept the trial-wise one as the HSSM contract and the shared one as the fast path
JAX / flax / orbax churn	0.5.3 vs Malte's 0.10.2 already needed two environments; orbax restore formats changed between them (MALTE_STAGES.md M4 part A)	pin by lock; the exported card is the durable artefact, the checkpoint is not; keep the NumPy port as the thing that reads weights
R version	R < 4.5 silently cannot load shipped artefacts	fix nle_file_hash; nle_verify prints the R version first
two copies of the Python code (flow/ and handover/nle-methods/flow/)	they have already diverged	the package replaces both; the handover copy becomes the examples
support burden	a likelihood that loads silently and fits silently (EMC2_IMPLEMENTATION_BRIEF.md:244-247)	the kit is the price of admission: register_nn_model() warns without a validation.sbc record

10.3 Maintenance

The Python package: Julian (author of flow/) with AH; it changes when a new model family needs a new structural option, which is rare. The EMC2 NN layer: whoever maintains dev-nle now (the EMC2 sessions under AH), inside EMC2's test suite, which is the right place because EMC2's samplers are what it is tested against. Malte's converter: Malte. If nobody will own the Python side, the honest alternative is option C (§6) with the scripts frozen at a tag and documented as "edit these six places": cheaper, and it is what exists now.

---

11. Recommendation, first milestone, decisions

11.1 The smallest useful first milestone (M0, about a week)

It is independent of the Python-package decision and it removes the three gaps found in this stage:

1. **Card spec v1**, written in EMC2 (vignettes/card-format.Rmd or inst/doc/CARD.md), with card_version, tail_mass, a structured scaling, and the validation block; nn_normalise_card() reads the version.

2. **`flow_scaled` in R**: `register_nn_model()` recognises a DDMnle card (four inputs, `scaling` present), builds the native spec the patch expects, and fills `fixed`/`refuse` from `DDM()`; `ddm_sc_native_reg()` and the scaffold go away; `DDMsc()`'s R path becomes the R-path twin of the compiled kind (like the other kinds). Acceptance: compiled vs `ddm_sc_density` ≤ 1e-11 on 200 draws × 6 models (the existing check in `scaled_cell.R`).
3. **`nn_golden_check(card, golden)`** in EMC2, used by the tests and by `nle_verify`.
4. **DDMnle golden vectors** for `ddm_sc_s1106_e245`, regenerated from the committed weights JSON by a script (`nleflow golden <card>`, the NumPy port's grid; §2.4.2), committed beside the card, plus the two runners-up if they are kept. The RDM card `rdm_ss_base_s801_e271` already has its golden file (§3.1) and uses it as is. No checkpoint access, no tux26.
5. **`nle_verify`** (§9.3) and the R < 4.5 fix.
6. The scaled patch pushed as an EMC2 branch (`dev-nle-scaled`) rather than a file in this repository.

11.2 Then, if the package is wanted

M1 Python package skeleton with the RDM example (cheap to train, exact control); M2 the DDMnle example and the ONNX graph; M3 converter move and presets; M4 docs, container, CI. Each milestone ends with `nle_verify` passing on a fresh machine.

11.3 Decisions AH needed to make next (all taken 2026-10-16; the answers are in §13)

1. **Build the Python package, or freeze the scripts** (option B vs C). Recommended B. **Decided: B.**
2. **No separate R package**: the R side lives in EMC2 `dev-nle`. **Decided: no separate R package.**
3. **Which RDM is the worked example** (§3.1). **Decided: the A = 0 box flow, `rdm_ss_base_s801_e271`.**
4. **Approve one tux26 export run** for the DDMnle golden vectors. **Approved, and then not needed** (§2.4.2): the golden vectors regenerate from the committed JSON. The approval stands unused unless the export format changes.
5. **The rebase of `dev-nle` onto `dev`**: before or after M0. **Recommended and decided: after** (§13.2).
6. The open conventions in §4.4. **Answered** (§13.3).

11.4 Deviation from the prompt

The prompt asks for the trailer `Co-Authored-By: Claude Sonnet 5.5`. This session ran on Fable 5.1, so the commit carries the trailer for the model that wrote it, plus the session link, rather than a false one.

---

12. Next stage (one model, one prompt)

Nothing blocks it (§13). Model: **Opus 5.5** (or Fable), auto mode; this is EMC2-side, trap-dense work with exact acceptance tests, the kind this project assigns to Opus/Fable (`handover/ZERO_BOX_RETRAIN_PLAN.md`, "which model for which stage"). It runs on the laptop, where the EMC2 checkouts and `~/R/nle-lib` are.

```
AUTO MODE REQUIRED: refuse to start unless this session is in auto mode.
Read handover/NLEFLOW_PLAN.md sections 1.1, 1.3, 2.4, 7.1, 9.3, 11.1 and 13 (nle repo, branch nleflow-plan), then
handover/nle-methods/port/R/ddm_scaled.R, handover/emc2/emc2_dev-nle-scaled.patch, and in the EMC2 dev-nle tree
R/nn_register.R (register_nn_model, nn_normalise_card, nn_native_spec, nle_file_hash), R/nn_validate.R
(nn_total_mass, nn_in_box), src/model_NN.h, tests/testthat/test-nn-port.R.
Do milestone M0 of the plan (section 11.1, all six items, as amended by section 13) on a local EMC2 branch
dev-nle-m0 off 142bd497 (the local dev-nle with the final sampler; the tarball in handover/emc2 is the same
tree),
installed into a NEW dedicated library; never into ~/R/nle-lib. Do not rebase onto dev in this stage (AH's
decision, section 13.2); push the scaled patch as the first commit of the branch so it stops being a file in
another repository.
Item 4 (DDMnle golden vectors) is a script, not a compute run: port the exporter's float64 NumPy path to a
standalone `golden.py <weights.json>` that reads the card alone (grid as in section 2.4.2), write
port/export/ddm_sc_s1106_e245_golden.csv with it, and check the compiled evaluator against it. Do not touch
tux26, Gadi or AWS, and do not regenerate anything from checkpoints.
Section 13.3 is part of the specification: the SZ contract (three conventions, one conversion function, a golden
row at each boundary); the s range and the v b / s^2 overflow limit recorded in race cards and checked at
```

```
registration; nn_in_box warns and never enforces; nn_total_mass integrates over the support the card states and
expects 1 within the card's tolerance; cap256w_c4's documentation says no t0 correction is applied; a warning
when rt_resolution is coarser than 1/1000 and t0 is sampled.
Acceptance, one line each: (a) register_nn_model("port/export/ddm_sc_s1106_e245_weights.json", twin = DDM)
returns a flow_scaled registration with no scaffold; (b) its compiled likelihood equals ddm_sc_density to <= 1e-
11
over 200 prior draws in each of the six regimes of sbc/scaled_cell.R; (c) nn_golden_check() passes on rdm_small,
cap256w_c4 and ddm_sc_s1106_e245 against their golden files and fails loudly on a perturbed golden row; (d) a
card
without card_version loads as version 0 and a card with card_version = 2 is refused by name; (e) on R < 4.5 the
shipped artefacts either load (MANIFEST carries md5) or fail with a message that names the R version; (f) the SZ
golden rows pass at all three boundaries; (g) Rscript inst/scripts/nle_verify.R prints one PASS line per card and
exits 0; (h) the full test-nn-* suite passes. The RDM card rdm_ss_base_s801_e271 (card, golden and intermediates
under handover/nle-methods/port/export/, sha256 b6ce7db9...) is part of (c) and (g): its golden file is the
reference; register it with twin = RDM, and do not regenerate it.
Work with absolute paths, at most 8 cores. Commit on the EMC2 branch and on nleflow-plan (the plan's progress
log, one line), push only those two branches, and end with ONE next stage: M1, the Python package skeleton with
the
RDM example (section 11.2), model Sonnet 5.5 or above, auto mode, with its full prompt.
```

13. Decisions taken (AH, 2026-10-16) and what they change

13.1 The five decisions

1. **Build the Python package** (option B, §6).
2. **No separate R package**; the R side lives in EMC2 `dev-nle`.
3. **The RDM worked example is the A = 0 box flow, `rdm_ss_base_s801_e271`** (§3.1), not Malte's. Its card, golden file and intermediates are on this branch (28020b7, committed by AH from the laptop's `port/export/`); the weights file's sha256 `b6ce7db9ca98ed17f1b04ecfc8860744fd5716ee20709f1baddcccdac0b44700` matches the Gadi bundles' record. The golden CSV is the RDM reference for the port tests. Final release location at packaging time.
4. **A tux26 export run was approved** for the DDMnle golden vectors, and is **not needed**: the probe in §2.4.2 shows the golden grid regenerates from the committed JSON through the exporter's own float64 port, and EMC2's compiled evaluator matches it to 1.2e-10. The existing exports under `handover/nle-methods/port/export/` are the package's reference files; a regeneration script ships with the package; nothing is regenerated from checkpoints unless the export format changes, and then only with AH's permission.
5. **Rebase `dev-nle` onto `dev` after M0, not before** (AH's preference; my recommendation agrees, §13.2).

13.2 Why the rebase can wait

Measured on the GitHub branches (2026-10-16): `dev` and `dev-nle` forked at a311c03e (14 Sep); since then `dev` has 5 commits and `dev-nle` 15 (plus the local sampler merge 142bd497 and the scaled patch). Files both branches changed: `NAMESPACE`, `NEWS.md`, `R/SBC.R`, `R/sampling.R`. The NN layer touches those two R files in one line each (`R/sampling.R:1010` on `dev-nle`, the `nn_native_args` hook; `R/SBC.R:34`, `nn_sbc_box_check`) plus the `qbinom` fix in `SBC.R` (154f21ad), while `dev` rewrote `sampling.R` (651 lines) and `SBC.R` (57) for the new sampler. `dev-sampler` is not an ancestor of `dev` (the sampler reached `dev` as 8b9bdbf, 6 Oct, by a different route), so the merge will have to reconcile the sampler twice, once for each side's copy, and that is work for whoever is still changing `dev`, when it is stable. M0 adds nothing to that merge: it changes `R/nn_register.R`, `R/nn_validate.R`, `src/model_NN.h`, `src/flow_ddm.cpp`, `src/nle_native.h`, `inst/scripts/`, `tests/testthat/test-nn-*` and a vignette, none of which `dev` touches. The risk of waiting is only that the hooks' line numbers move; the usability gate (`EMC2_IMPLEMENTATION_BRIEF.md:301-351`, 25 min at 8 cores) is the check after the rebase, whenever it happens. So: M0 now on `dev-nle`; rebase once, after AH's `dev` work lands, before M1's EMC2 side needs it.

13.3 The conventions, now contracts (AH's detailed answers)

1. **SZ.** Three conventions exist and each is defined once in the card schema with a conversion function and a golden row at its boundary: the network's SZ is *relative* (SZ/m, m = min(Z, 1 − Z); `ddm_scaled.R:30-35`); EMC2's `DDM()` passes `sw = 2 SZ min(Z, 1 - Z)` after `Ttransform`; the scaffold passes a proportion (`scaled_cell.R:53-57`). M0's `flow_scaled` kind carries the conversion and the golden rows (acceptance (f) in §12).
2. **s = 1 and the ±4 sd rule** are cell conventions, not package defaults. Race cards record the s range and the v b / s^2 limit (pWald overflows above 354, `build_rdm_cell.R:12-13`) and registration checks them; the prior-inside-box rule becomes a validation helper (`nn_in_box`) that warns and never enforces.
3. **`--affine` and `--decision-time`** are experiment options: optional training flags, the baseline (neither) as the default, documented as unproven, revisited when the seed-study reading arrives.
4. **Tail mass.** `nn_total_mass()` integrates over the time support the card states and expects 1 within the tolerance the card states; the card records how mass beyond the support is treated (for DDMnle: the identity tails beyond the spline range [−12, 2] on log u carry the remainder, so the density integrates to 1 over u ∈ (0, ∞) and nothing is lost, §2.4.4). This replaces the `tail_mass` field proposed in §7.1 with three: `support` (the time range the card is valid on; `(0, Inf)` for DDMnle), `mass_tolerance`, and a `beyond_support` note. Implemented in M0 (`nn_total_mass()` reads them; `vignette("neural-likelihood-cards")`).
5. **The t0 correction** is not carried by the package. The fixed delta 0.00276 was reopened on 2026-10-13 (exact test: mean rank .534, KS p .007; an out-of-sample test of about 0.0019 pending; `OPEN_ITEMS.md:31-35`). `cap256w_c4`'s documentation says that no correction is applied, that an unresolved t0 shift of about 0.002 at N = 800 remains, that DDMnle replaces it for the scaled cases, and links the exact-test record.
6. **`rt_resolution ≤ 1/1000`**: a vignette note plus a warning when the data resolution is coarser and t0 is sampled (EMC2's default 1/60 s flooring biases t0 by about −1.9 posterior sd at N = 100 even for the analytic likelihood, `METHODS.md:714-715`).
7. **Golden vectors** are a fixed set of inputs with the outputs the network must return, computed once from the trained weights; every port (compiled EMC2, R reference, ONNX, any future PyTorch port) is tested against them. The existing exports under `handover/nle-methods/port/export/` are the reference files; the package includes the regeneration script (§2.4.2); no regeneration from the tux26 checkpoints without AH's explicit permission, and only if the export format changes.

Progress log

- 2026-10-16 plan written (this file); probes run on the cloud instance as in §1.3; nothing on tux26, Gadi or AWS; no push to ampl-psych/EMC2.
- 2026-10-16 AH's decisions recorded (§13); §2.4.2, §3.1, §11.1, §11.3 and the §12 prompt amended. Probe: DDMnle golden vectors regenerated from the committed JSON match EMC2's compiled evaluator to 1.2e-10 (log pdf) and 6e-15 (CDF) over 1150 rows, so the approved tux26 export is not needed.
- 2026-10-16 the RDM A = 0 card committed by AH (28020b7). Verified here: weights sha256 `b6ce7db9ca98ed17f1b04ecfc8860744fd5716ee20709f1baddcccdac0b44700` = the Gadi bundles' record; card fields `context_names v,B,t0,s,A`, `A: zerobox` with c = 0.05, box A ∈ [0, 2], [256, 128], 64 bins, range ±5. R reference port (`port/R/flow_race.R`) vs its golden CSV: 575 rows, max |Δlog pdf| 9.6e-13, |ΔCDF| 3.6e-15, |Δlog sf| 3.1e-14, 115 rows at exactly A = 0. `register_nn_model(card, twin = RDM)` in dev-nle 142bd497 registers it as `flow_race`, `c_name NN`; the R-path evaluator vs golden: |Δlog pdf| 1.9e-12, |Δlog sf| 6.1e-14. The "card not in the repository" case is withdrawn from §3.1, §11.1, §12 and §13.
- 2026-10-16 **M0 done** (cloud instance, R 4.3.3; EMC2 branch `dev-nle-m0`: the scaled patch as its first commit f7aa07ea, then the 142bd497 tree, then the M0 commit 39ae8d8a. **Correction 2026-10-17:** the push to ampl-psych/EMC2 was refused (HTTP 403, Claude GitHub App not installed for the organisation), so the branch travelled as `handover/emc2/dev-nle-m0/000*.patch`; AH applied the three patches on the laptop on 2026-10-17 (clean, built into `~/R/nle-m0-lib`, verification passed), pushed `dev-nle-m0` to ampl-psych/EMC2 from there and removed the patch files (f165e6c). This repo: `golden.py`, `ddm_sc_s1106_e245_golden.csv`, `scaled_cell.R` M0 mode). Acceptance lines, as run:
    a. `register_nn_model("port/export/ddm_sc_s1106_e245_weights.json", twin = DDM)` returns a `flow_scaled` registration, no scaffold (`sbc/scaled_cell.R` with `M0=1`; `test-nn-scaled.R`);

   b. compiled vs `ddm_sc_density` over 200 prior draws, six regimes: full 7.0e-13, sv0 1.4e-12, SZ0 9.1e-13, st00 2.3e-12, st0only 1.4e-12, SZonly 1.5e-12 (≤ 1e-11); compiled DDMnle 0.35–0.54 ms per likelihood here against WienR, 1.6–5.5x;
   c. `nn_golden_check()` PASS on rdm_small 9.2e-12, cap256w_c4 (flow part) 3.1e-12, st0zero 1.4e-11, ddm_sc_s1106_e245 8.5e-11 (its golden file regenerated from the JSON by `golden.py`; the script reproduces the committed goldens of rdm_small, rdm_ss_base_s801_e271 and z0_s503_e282 to 1e-12 and cap256w's flow part to 2.5e-13), rdm_ss_base_s801_e271 2.1e-12; a perturbed row (1e-6) fails by name;
   d. a card without `card_version` loads as 0, `card_version = 2` is refused as "Card version 2 is not supported";
   e. on R 4.3.3 the shipped artefacts load through the MANIFEST's md5 lines (before M0 all three failed there);
   f. the SZ conventions meet at the three boundaries (DDM()'s Ttransform ↔ `nn_sz_convert` exact to 1e-15; the registered model's proportion vs the reference's width ≤ 1e-12);
   g. `inst/scripts/nle_verify.R` prints PASS on every line (three shipped artefacts, DDMnle golden, compiled vs R path 2.1e-13, oracle 8.3e-14, the RDM card) and exits 0;
   h. `test-nn-*`: 919 expectations, 0 failures, 0 errors, 5 skips (python/onnxruntime and reg_s402 env vars), 0 warnings. Not done: the rebase onto `dev` (AH's decision); nothing on tux26, Gadi or AWS.
- 2026-10-17 **M1 done** (cloud instance, 4 cores, R 4.3.3 with EMC2 dev-nle-m0 in a scratch library; this repo only, `nleflow/` on nleflow-plan). The package: `pyproject.toml` + `uv.lock` (jax 0.5.3, flax 0.10.6, distrax 0.1.5, optax 0.2.8, orbax-checkpoint 0.11.1, numpy 2.2.6, scipy 1.15.3, pyarrow, pyyaml, onnx 1.23.2, onnxruntime 1.31.0; pytest in the dev group); `src/nleflow/spec.py` (section 7.2's schema, validated by name, defaults filled; `response: choice` and `scaling` refused as M2), `data.py` (the parquet contract, a CSV fallback for R without arrow, the zero-box encoding in float64, the standardiser as `scaler.json`), `train.py` (train_race.py's loss, AdamW warmup-cosine, Orbax checkpoints), `export.py` (card_version 1 race cards with `support`, `mass_tolerance`, `beyond_support`, `control_limits`; the float64 JAX gate at 1e-8 replaces export_flow.py's float32-only gate, the float32 gap printed), `golden.py` (M0's script moved in, `nleflow golden`; the handover script is now a shim), `simulate.py`, `cli.py` (spec, simulate, train, steps, export, golden); `flow/` vendors config.py and checkpoints.py unchanged; `examples/rdm_a0/{spec.yaml, spec_smoke.yaml, simulate.R, README.md}` (the simulator is sim_rdm_a0.R driven by the spec); `tests/` (13 pytest cases); `README.md` with section 9.2's Python steps, each with its check. Acceptance, as run: (a) `uv sync --frozen --group dev` from the lock (fresh `.venv`, 11 s warm), then `nleflow --help` lists the six commands; (b) the smoke spec: 240 000 + 24 000 rows simulated in 5 s (CSV: no `r-cran-arrow` on this Ubuntu), 2 epochs in 13 s (val loss 0.664), export in 15 s; the card registers in EMC2 with `twin = RDM` as `flow_race`, card_version 1, `support (0, Inf)`, `control_limits` read back, and `nn_golden_check()` passes at 2.7e-13; (c) `nleflow golden` reproduces `rdm_ss_base_s801_e271_golden.csv` to 3.0e-13 and `rdm_rdm_small_e195`'s golden to 5.2e-14 (that file copied from EMC2's test tree to `port/export/` as the reference beside its card); (d) the float64 gate on the smoke card: max |dlog pdf| 2.0e-13, |dz| 7.4e-15 against 1e-8; float32 gap 3.6e-5 printed; (e) `nle_verify.R <smoke card>`: every line PASS, exit 0;
   f. `pytest`: 13 passed (the 13th registers a synthetic card in R when `NLEFLOW_EMC2_LIB` is set). Two deviations, both deliberate: the card carries a `validation` block only once it records something (an empty block would silence EMC2's "records no validation" message); the spec separates `training.seed` (the network) from the simulator's data seed (`NLEFLOW_SIM_SEED`, default 4876). Found and fixed: M0's `golden.py` read `pre_bounds_natural` as a list where the exporters write `{lower, upper}` (decision-time cards only). Not done: no full training, nothing on tux26, Gadi or AWS, no change to EMC2.
- 2026-10-17 **M2 done** (cloud instance, 4 cores, R 4.3.3 with EMC2 dev-nle-m0 in a scratch library; this repo only, `nleflow/` on nleflow-plan; nothing on tux26, Gadi or AWS; DDMnle not retrained; no change to EMC2). The choice-model path: `spec.py` grows `response: choice`, `classifier`, `scaling {kind: ddm_unit, sz: proportion}`, long-form sampling strata (`uniform`, `log_uniform`, `asinh_uniform`, `zero`, each with a range) and `regimes` (joint exact zeros); `train.py` conditions the flow on (context, R) and trains the classifier afterwards on the exact P(R = 2) sets with retrain_classifier.py's tuned recipe (`classifier.json`; `--classifier-from` reuses one when the standardisers agree to 1e-3); `export.py` writes flow_joint / flow_scaled cards (version 1: the structured `scaling`, `context_encoding zerobox`, `classifier_mlp`, `classifier_scaler`, `support (0, Inf)` of u) and gates the joint log density of both responses against JAX

float64 at 1e-8; `golden.py` returns the joint arrays and intermediates; `onnx_export.py` + `nleflow onnx` / `onnx-check`: the LAN-shaped graph of a flow_joint or flow_scaled card in the trialwise `[N, k + 2]` and shared (theta, rt, choice) signatures, float64, opset 18, IR 10, the scaled graph taking the natural DDM parameters and doing kappa, eta, the encodings, c, u, the st0 branch (density at the window midpoint when st0 = 0 or st0 < 1e-7 d, else [G(u1) - G(max(u0, 0))] / st0) and the rt <= t0 rule inside, exactly nn_eval_scaled(); onnxruntime has no double Erf or Asinh, so the graph carries R's pnorm (Cody) and qnorm (AS 241) as graph operations (`rmath.py`, vs scipy 2e-14 / 2e-15) and asinh as log(x + sqrt(x^2 + 1)); `scripts/onnx_cases_scaled.R` writes EMC2's reference cases; the DDMnle example (`examples/ddmnle/`: spec, 1 % / 2-epoch smoke spec, the spec-driven scaled simulator with WienR targets, README); nine new pytest cases; READMEs. Acceptance, as run: (a) the smoke DDMnle spec: 240 000 + 10 000 rows and 2000 + 200 choice targets simulated in 10 s, flow 2 epochs in 11 s (val loss -1.107; the project's 300-epoch flow -1.436), classifier 194 epochs in 6 s (val mean |P_hat - P| 0.0046), export in 27 s; the card registers with `twin = DDM` as flow_scaled, card_version 1, `scaling ddm_unit`, and `nn_golden_check()` passes at 3.6e-13;

b. `nleflow golden` reproduces `ddm_sc_s1106_e245_golden.csv` exactly (max |diff| 0, 1150 rows); (c) float64 gate on the smoke export 3.8e-13 (log pdf) and 1.1e-14 (z), float32 gap 7.5e-5 printed; (d) the ONNX graphs of the committed card, both signatures: vs the golden rows max |dlog p| 5.3e-11 over 1010 rows above the floor (median 7e-15; the 12 rows above 1e-11 sit at z = 2.0, the spline's range boundary), vs EMC2's nn_eval_scaled over 200 prior draws x 100 trials in the six regimes (half the draws with s != 1): 4.4e-11 where the likelihood is a density (st0 = 0; 38 400 rows) and 5.6e-9 on the st0 > 0 branch (78 400 rows, median 4e-15), the -Inf rows below t0 and the floor rows agreeing; the NumPy float64 port itself differs from EMC2 by 3e-9 on the st0 > 0 rows, so the 1e-11 target is met where the formula is a density and is limited by the conditioning of the CDF difference G(u1) - G(u0), not by the graph, where it is not (`onnx-check` defaults: 1e-10 and 1e-8); the eight-parameter graph of the st0zero card z0_s503_e282 vs its golden rows 1.3e-12; (e) `nle_verify.R <smoke scaled card>`: golden PASS, compiled vs R path 5.7e-14 PASS, oracle 8.3e-14 PASS (with EMC2_NLE_PORT), exit 0; (f) pytest: 25 cases, all pass with `NLEFLOW_EMC2_LIB` (3 of them need it, 1 the EMC2 checkout's golden file). Deviations: the ONNX agreement on the st0 > 0 branch (above); the simulator draws pseudo-random uniforms where the project drew shifted Sobol points; the classifier is trained with the run rather than copied from a checkpoint (the project's `--classifier-from` becomes `nleflow train --classifier-from <run>`).

Next stage after M0: M1, the Python package skeleton with the RDM example

Model: **Sonnet 5.5** (or above), auto mode. Laptop or cloud instance; needs Python 3.11/3.12, `uv`, and R with EMC2 `dev-nle-m0` in a dedicated library for the acceptance lines that touch EMC2.

```
AUTO MODE REQUIRED: refuse to start unless this session is in auto mode.
Read handover/NLEFLOW_PLAN.md sections 0, 3, 4, 5, 7.2, 8, 11.2 and 13 and the M0 entry of its progress log (nle
repo,
branch nleflow-plan); then handover/nle-methods/flow/scripts/train_race.py, export_flow.py, golden.py,
handover/nle-methods/flow/src/flow/{config,data,training,checkpoints}.py, handover/nle-
methods/simulation/sim_rdm_a0.R
and sim_functions.R, and EMC2's vignette neural-likelihood-cards.Rmd (branch dev-nle-m0). Do milestone M1: the
Python
package `nleflow` under nleflow/ in this repository (pyproject + uv lock pinning jax 0.5.3, flax 0.10.6, distrax
0.1.5, optax 0.2.8, orbax-checkpoint < 0.11.2, numpy 2.2.6, scipy 1.15.3, plus onnx and onnxruntime), with:
  1. a model specification (section 7.2's YAML/JSON: model, response race|choice, context, transforms, zerobox_c,
     box_natural, sampling strata, pre, scaling, spline, conditioner, classifier, training, simulator command);
  2. one `nleflow train <spec>` command that replaces train_race.py for race specs (train_ddm.py's DDM path is
M2),
     reading the parquet contract of section 7.2 and writing checkpoints under a run directory named by the spec;
  3. one `nleflow export <run> --step N` command that writes a card_version = 1 card (EMC2's vignette is the
spec:
     context, transforms, zerobox_c, bounds, scaler, mlp, spline, pre, support, mass_tolerance, beyond_support,
     control_limits for race cards, an empty validation block) with the float64 gate against JAX float64 at 1e-8
```

```
     (export_flow.py gates against float32 only: change that) and its golden file via golden.py's code, moved
into
     the package as `nleflow golden <card>`;
  4. the RDM A = 0 worked example as a spec (context v, B, t0, s, A; A zerobox c = 0.05; the three-part A
mixture;
     [256, 128], 64 bins, range +-5; batch 1024; 300 epochs; 30 kept checkpoints) with a 1 % smoke configuration
     (2 epochs) that runs end to end on a laptop in minutes;
  5. a regression test that `nleflow golden` reproduces port/export/rdm_ss_base_s801_e271_golden.csv and
     rdm_rdm_small_e195's golden to 1e-12 from their JSON cards (golden.py does today), and that a smoke-trained
     card registers in EMC2 dev-nle-m0 with twin = RDM and passes nn_golden_check() on its own golden file;
  6. a README with the install steps of section 9.2 for the Python side, each with its one-line check.
Do not change the exporter's card fields beyond what the vignette names; do not regenerate anything from
checkpoints; do not touch tux26, Gadi or AWS; no DDM training (M2). Keep --affine and --decision-time as optional
flags, off by default, documented as unproven (section 13.3.3).
Acceptance, one line each: (a) `uv sync` from the lock on a fresh machine, then `nleflow --help`; (b) the smoke
spec trains 2 epochs on 1 % data and exports a card that registers in EMC2 and passes nn_golden_check(); (c)
`nleflow golden` reproduces the two committed RDM goldens to 1e-12; (d) the exported card's float64 gate passes
at
1e-8 and the float32 gap is printed; (e) `Rscript inst/scripts/nle_verify.R <smoke card>` prints PASS lines and
exits 0; (f) pytest passes.
Work with absolute paths, at most 8 cores. Commit on nleflow-plan only (this repository), push only that branch,
append one line to the plan's progress log, and end with ONE next stage: M2 (the DDMnle example and its ONNX
graph
with the CDF, preprocessing and st0 branch, section 2.4.1), model Opus 5.5 or Fable, auto mode, with its full
prompt.
```

Next stage after M1: M2, the DDMnle example and its ONNX graph

Model: **Opus 5.5** (or Fable), auto mode. Laptop or cloud instance; needs the `nleflow` environment (`uv sync --frozen --group dev` in `nleflow/`) and R with EMC2 `dev-nle-m0` in a dedicated library. The scaled DDM is the trap-dense half of the package (three SZ conventions, the st0 branch, exact zeros, a float64 ONNX graph with a CDF), which is why this stage goes to Opus/Fable.

```
AUTO MODE REQUIRED: refuse to start unless this session is in auto mode.
Read handover/NLEFLOW_PLAN.md sections 2, 4, 7.1-7.3, 7.5 and 13.3 and the M0 and M1 entries of its progress log
(nle repo, branch nleflow-plan); nleflow/README.md and nleflow/src/nleflow/{spec,data,train,export,golden}.py;
handover/nle-methods/flow/scripts/{train_ddm.py,export_flow_ddm.py,export_onnx_ddm.py,onnx_check.py},
handover/nle-methods/simulation/{scaled.R,zerobox.R}, handover/nle-methods/port/R/ddm_scaled.R and
port/tests/test_ddm_scaled.R, handover/SCALED_DDM_PLAN.md, handover/REPORT_RUSS_JULIAN.md (the ONNX section); and
on EMC2 dev-nle-m0: vignettes/neural-likelihood-cards.Rmd, R/nn_register.R (nn_eval_scaled, nn_scaled_inputs,
nn_sz_convert), src/flow_ddm.cpp (nn_trial_ll_scaled), tests/testthat/test-nn-scaled.R,
inst/scripts/nle_verify.R.
Do milestone M2: the scaled DDM (DDMnle) path of nleflow and its ONNX export.
  1. The spec grows `response: choice` with `classifier: {widths: [...]}` and `scaling: {kind: ddm_unit}`:
     context kappa, eta, Z, SZ (eta and SZ zerobox with c = 0.01 and 0.05, stored log and probit; kappa identity;
     Z probit), spline range [-12, 2], the box of export_flow_ddm.py's scaled registry, and the SZ convention
     recorded as `proportion` (section 13.3.1). Keep the race path untouched; a spec that mixes the two is
refused.
  2. `nleflow train` for choice specs = train_ddm.py --scaled: the flow conditioned on (context, R) and the
choice
     classifier on the context, trained together or the classifier reused from another run (train_ddm.py's
     --classifier-from); the parquet contract of section 7.2 with R; read how train_ddm.py builds the
classifier's
     target before writing anything.
  3. `nleflow export` for flow_scaled cards: card_version 1 with the structured `scaling` block,
`context_encoding
     zerobox`, `support {time: [0, "Inf"]}` of the unit-scale time, the classifier MLP, the float64 gate at 1e-8
     (export_flow_ddm.py already gates against JAX float64; carry it) and the golden file from golden_joint().
  4. The DDMnle example: nleflow/examples/ddmnle/{spec.yaml, spec_smoke.yaml, simulate.R, README.md}, the
     simulator simulation/scaled.R driven by the spec (a = s = 1, st0 = 0, the zero-box strata of zerobox.R), the
     recipe of ddm_sc_s1106_e245 (SCALED_DDM_PLAN.md), and a 1 % / 2-epoch smoke configuration. DDMnle itself is
     NOT retrained: the committed card ddm_sc_s1106_e245 and its M0 golden file are the references.
  5. `nleflow onnx <card> [--signature trialwise|shared]`: the ONNX graph of a flow_scaled card (section 2.4.1):
     inputs the natural DDM parameters (probit and log stay with the caller), rt and the choice; inside the graph
```

```
     kappa, eta, c = (s/a)^2, u = (rt - t0) c, the st0 = 0 branch c g(u) and the st0 > 0 branch [G(u1) -
G(max(u0,
     0))] / st0 with the spline CDF and the normal CDF through Erf, zero for rt <= t0; float64, opset 18, IR 10;
     both signatures of section 7.3. Keep export_onnx_ddm.py's 8-parameter graph as the plain flow_joint case.
  6. Tests: `nleflow golden` reproduces port/export/ddm_sc_s1106_e245_golden.csv to 1e-12; the smoke scaled card
     registers in EMC2 with twin = DDM and passes nn_golden_check() and nle_verify.R's scaled checks (compiled vs
R
     path, the oracle); the ONNX graph of the committed card agrees with its golden rows and with EMC2's compiled
     likelihood (nn_eval_scaled over 200 prior draws in the six regimes of sbc/scaled_cell.R) to 1e-11 in
     onnxruntime, both signatures; the README gains the onnx steps and their checks.
Do not change the card fields beyond what the vignette names (an ONNX provenance field is fine); do not
regenerate anything from checkpoints; do not retrain DDMnle; do not touch tux26, Gadi or AWS; no push to
ampl-psych/EMC2 and no push to main. Should EMC2 need a change (none is expected), it goes as a patch under
handover/emc2/ with a note in the log, as M0's did.
Acceptance, one line each: (a) a choice spec loads and the smoke DDMnle spec trains 2 epochs on 1 % data and
exports a flow_scaled card_version 1 card that registers with twin = DDM and passes nn_golden_check(); (b)
`nleflow golden` reproduces ddm_sc_s1106_e245_golden.csv to 1e-12; (c) the float64 gate passes at 1e-8 on the
smoke export and the float32 gap is printed; (d) `nleflow onnx` on the committed card: onnxruntime vs the golden
rows and vs the compiled EMC2 likelihood <= 1e-11, both signatures, with the st0 > 0 and st0 = 0 regimes and
exact zeros of sv and SZ covered; (e) `Rscript inst/scripts/nle_verify.R <smoke scaled card>` prints PASS on
every
line and exits 0; (f) pytest passes.
Work with absolute paths, at most 8 cores. Commit on nleflow-plan only (this repository), push only that branch,
append one line to the plan's progress log, and end with ONE next stage: M3 (the ONNX -> card converter moved
into
the package with batched inputs, Relu/Gelu and presets, section 7.4; model Opus 5.5 or Fable, auto mode), with
its
full prompt.
```

Next stage after M2: M3, the ONNX → card converter moved into the package, with presets

Model: **Opus 5.5** (or Fable), auto mode. Laptop or cloud instance; needs the `nleflow` environment and R with EMC2 `dev-nle-m0` in a dedicated library. This is the stage that makes the published LAN library usable from EMC2 designs (plan section 7.4), and the stage where the converter's refusals must name what they refuse.

```
AUTO MODE REQUIRED: refuse to start unless this session is in auto mode.
Read handover/NLEFLOW_PLAN.md sections 7.1, 7.3, 7.4, 7.5 and 8 and the M0, M1 and M2 entries of its progress log
(nle repo, branch nleflow-plan); nleflow/README.md and nleflow/src/nleflow/{onnx_export,golden,spec}.py; on EMC2
dev-nle-m0: inst/scripts/onnx_to_card.py, R/nn_register.R (the mlp_joint kind: input_layout, response_values,
output_scaler, ll_floor_log), src/mlp_lik.cpp, tests/testthat/test-nn-mlp.R (registers lan_ddm_uniform_st),
tests/testthat/golden/{ddm_uniform_st.onnx, lan_ddm_uniform_st_ort.csv}, and
handover/nle-methods/port/export/lan_ddm_uniform_st.json; handover/EMC2_IMPLEMENTATION_BRIEF.md (the HSSM
batching
note) and REPORT_RUSS_JULIAN.md Appendix A.1 (the LAN parameter mapping).
Do milestone M3: the ONNX -> card converter in nleflow.
  1. `nleflow convert <model.onnx> --preset <name> [--out card.json]`: onnx_to_card.py's graph reader moved into
     the package (one non-initializer input, MatMul+Add or Gemm, Tanh; now also Relu and Gelu, and a batched
     [N, k] input as HSSM's converted graphs have), writing a card_version 1 mlp_joint card with input_layout,
     response_values, output_scaler, ll_floor_log, the preset's parameter mapping and box, and
     `validation: {calibration: "unknown; needs its own SBC cell before inference"}`. Each new activation is also
     a C++ change in EMC2's load_mlp (nle_flow.h) and that is out of scope here: convert with the activations
EMC2
     evaluates, and REFUSE the others with a message naming the first unsupported node, its op type and its name.
  2. Presets: one per published LAN the project has used (ddm_uniform_st today: a_EMC2 = 2 a_LAN, Z = z,
     t0 = t - st, st0 = 2 st, sv = SZ = 0, s = 1, box from the HSSM model config), as data (YAML/JSON under
     nleflow/presets/), with the mapping and its source cited, and `nleflow convert --list-presets`.
  3. `nleflow convert --verify`: the card's evaluation (nleflow's own NumPy MLP) against onnxruntime on the
card's
     box, rank-1 and batched inputs, to 1e-6 (float32 sources), writing the ort CSV the EMC2 test tree holds
     (lan_ddm_uniform_st_ort.csv's format) as the card's golden file.
  4. Regression tests: converting tests/testthat/golden/ddm_uniform_st.onnx reproduces the fields of
     lan_ddm_uniform_st.json (the committed conversion) and its ort CSV to 1e-6; the converted card registers in
     EMC2 dev-nle-m0 as mlp_joint and passes nn_golden_check() against that CSV at the float32 tolerance; a graph
```

```
     with an unsupported op is refused by name; README section with the install step (onnx, onnxruntime are in
     the lock) and the one-line check.
  5. Retire inst/scripts/onnx_to_card.py's duplicate: not by deleting it from EMC2 (no EMC2 change in this stage)
     but by a note in nleflow's README that nleflow convert is the maintained one, and a line in the plan.
Do not change the card fields beyond what the vignette names; do not touch tux26, Gadi or AWS; no push to
ampl-psych/EMC2 and no push to main; no SBC runs (the LAN's SBC is pending separately, REPORT_RUSS_JULIAN.md
A.3).
Acceptance, one line each: (a) `nleflow convert tests/.../ddm_uniform_st.onnx --preset ddm_uniform_st` writes a
card whose fields equal the committed lan_ddm_uniform_st.json up to provenance; (b) --verify agrees with
onnxruntime to 1e-6 on rank-1 and batched inputs; (c) the card registers in EMC2 as mlp_joint and passes
nn_golden_check() against the ort CSV; (d) a graph with a Sigmoid node is refused with a message naming the node;
(e) pytest passes; (f) README updated with the convert steps and checks.
Work with absolute paths, at most 8 cores. Commit on nleflow-plan only (this repository), push only that branch,
append one line to the plan's progress log, and end with ONE next stage: M4 (docs, container, CI, the rebase of
dev-nle onto dev once AH's dev work settles; model Sonnet 5.5 or above, auto mode), with its full prompt.
```